

Lesson 2 Action Group Programming

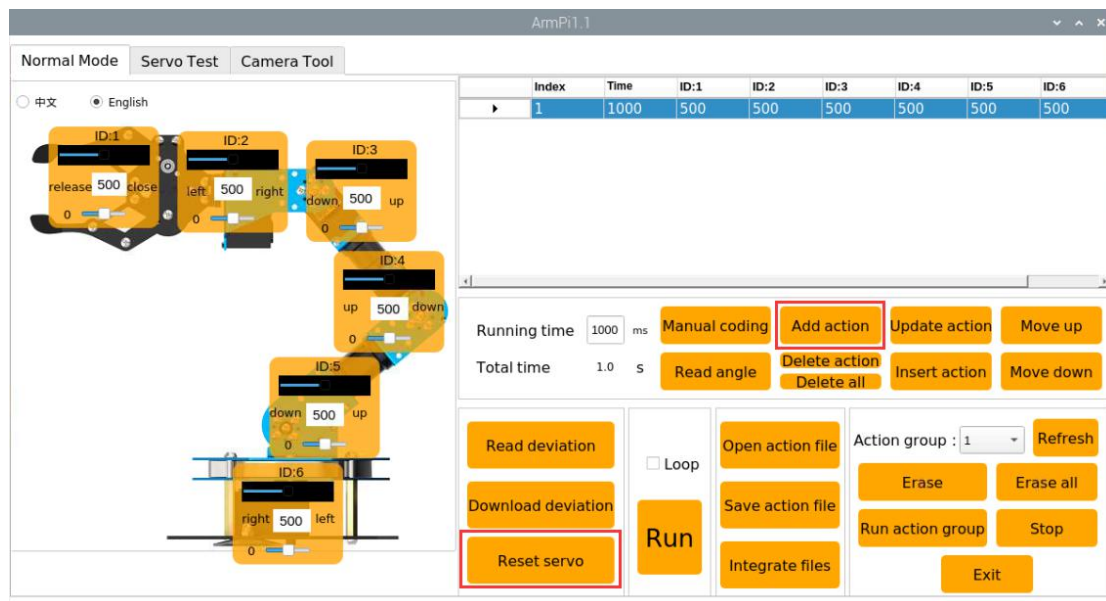
1. Project Outcome

Program an action group consisting of 7 actions to allow the ArmPi to "Grip".

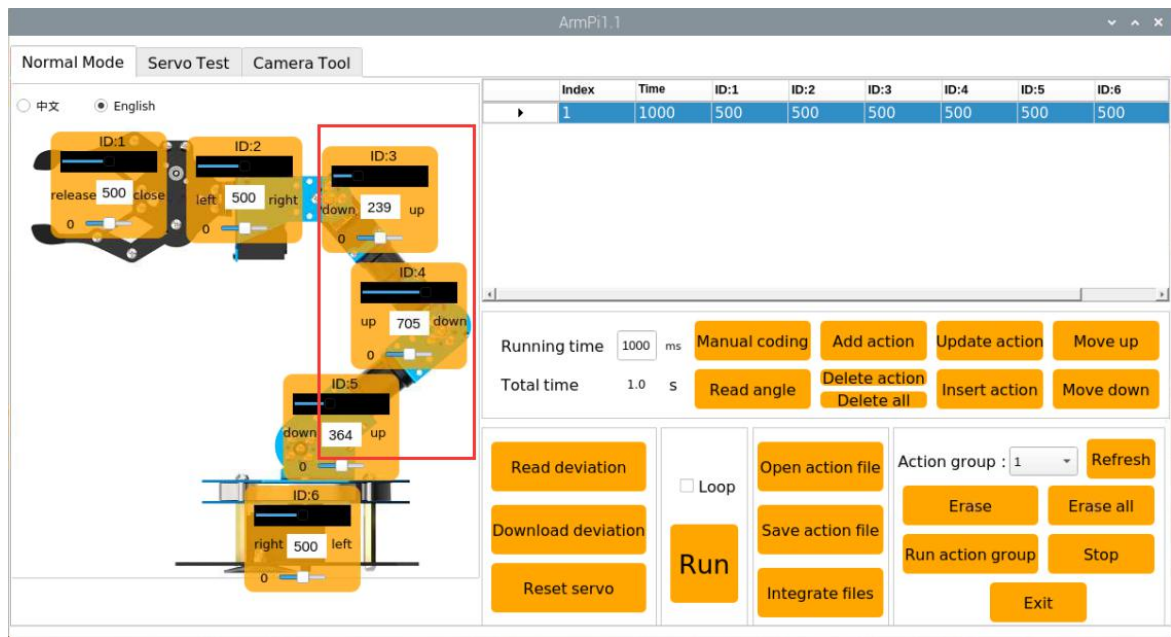
2. Complete Program

2.1 Action Programming

Step 1: Click "Reset Servo" to command all the servos back to the middle position (initial position), then click "Add action" to add this action to the edited list.



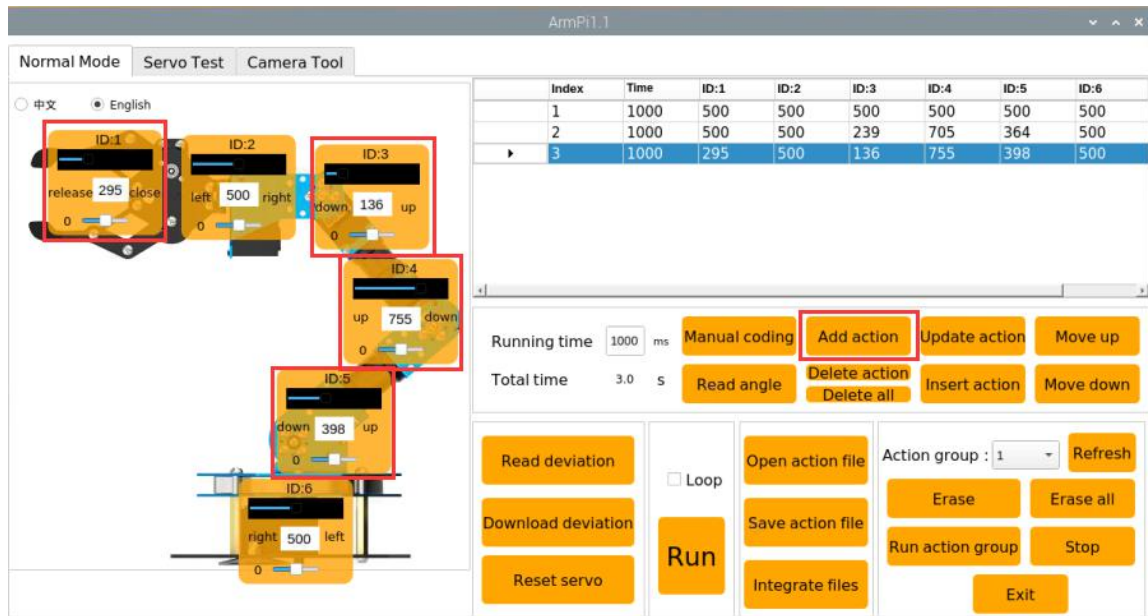
Step 2: Adjust to bend the arm. Drag the sliders of ID3, ID4 and ID5 to the value below



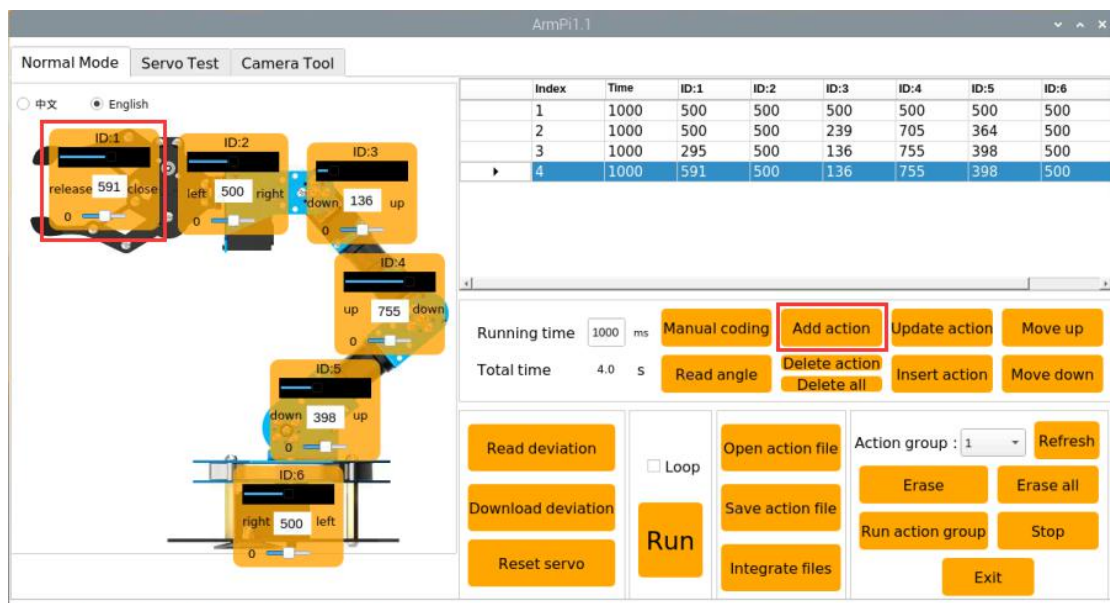
Step 3: Click “Add action” to add the second action.



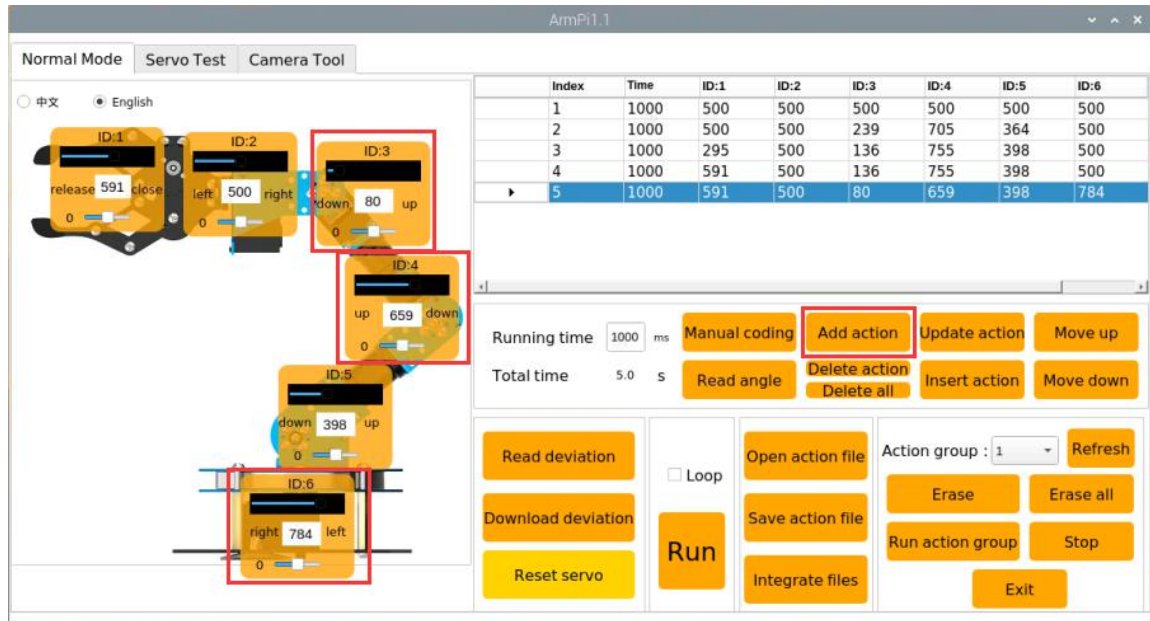
Step 4: Continue to lower the arm and open the gripper. Drag the sliders of the No.1, 3, 4, 5 servos as the figure below. Then click “Add action” to add the third action.



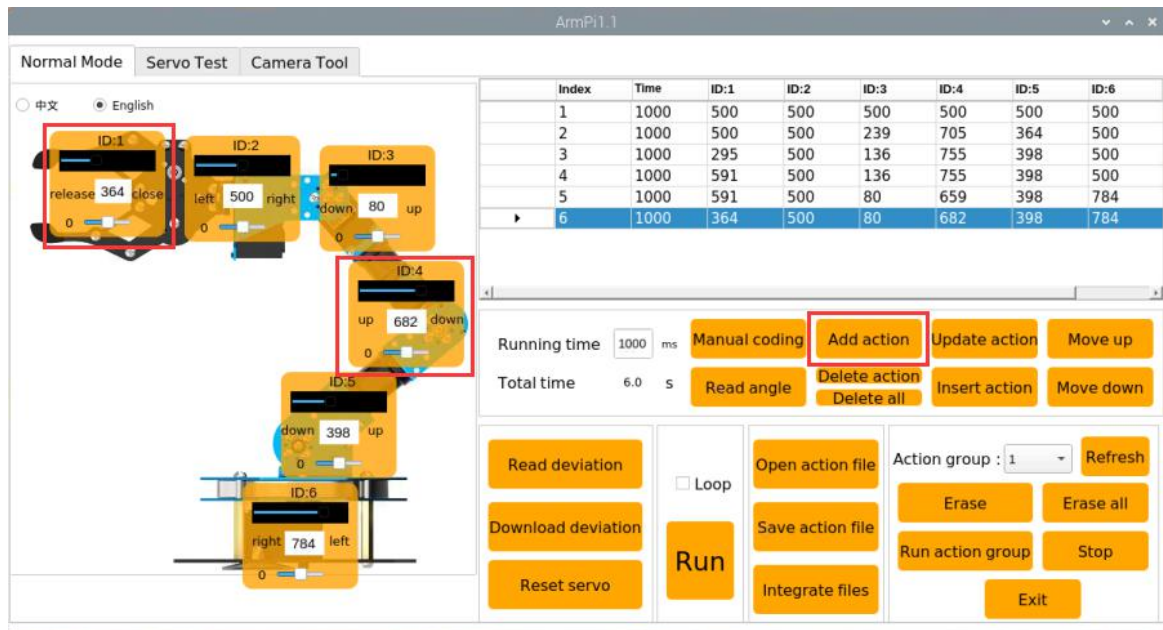
Step 5: Close the gripper to grab the object. Drag the ID1 to 591 then add it to the fourth action.



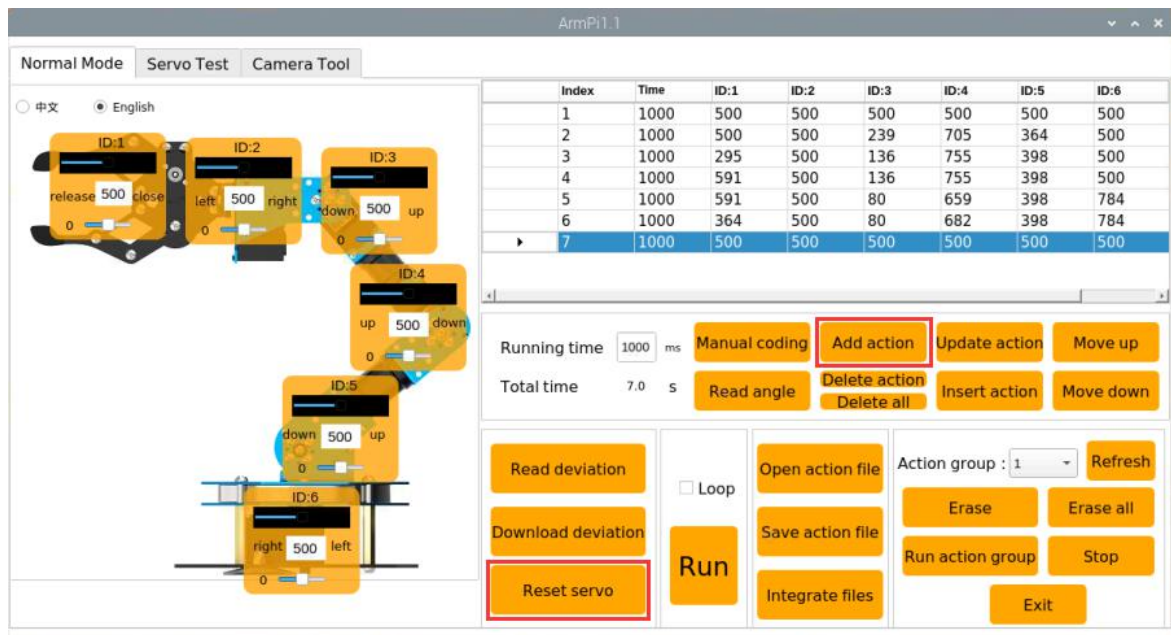
Step 6: Raise the robotic arm and rotate the base to the left, as shown below:



Step 7: Drag to put down the object, as shown below:



Step 8: Set the ArmPi to the initial position. Click “Rest servo” and “Add action”.



2.2 Save Action Group

After programming, please save the action group for future debugging. Click “Save action file”, choose the path as “/home/pi/ArmPi/ActionGroups”, name it (such as “3-pick up”) and then click “Save” button.

